

Fakultät für
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Towards Trustworthiness in AI-based Big Data Analysis

Lehrgebiet Multimedia und Internetanwendungen
Forschungsbericht

Towards Trustworthiness in AI-based Big Data Analysis

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April 26, 2024

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Abstract. The application of artificial intelligence has become extremely popular nowadays. Whereas “*AI is a foundational catalyst for digital business*”, in reality most organizations struggle to scale their AI pilots into enterprise-wide production, which limits the ability to realize AI’s potential business value. To be more precise, organizations “*are discovering ‘AI pilot paradox’, where launching pilots is deceptively easy but deploying them into production is notoriously challenging*”. One reason for this situation is “*that the risks and challenges that AI poses, including codifying and reinforcing unfair biases and infringing on human rights such as privacy, are undeniable*”. Thus, these issues are causing public concern and raising questions about the trustworthiness and regulation of AI applications. “*If AI systems do not prove to be worthy of trust, their widespread acceptance and adoption will be hindered, and the potentially vast societal and economic benefits will not be fully realized*”.

Related to the overall goal of this research (*Trustworthiness in AI-based Big Data Analysis*), this paper is focusing on the definition of intelligence, artificial intelligence, trust, and trustworthiness first. Afterwards and to integrate trustworthy artificial intelligence into the whole big data analysis process, relevant related work will be outlined and discussed. Based on these results, a conceptual reference model for trustworthiness in AI-based big data analysis (TAI-BDM Reference Model) will be introduced and discussed.

Keywords: Data Science · Artificial Intelligence · Trustworthiness · TAI-BDM Reference Model

1 Introduction

The application of *artificial intelligence (AI)* has become extremely popular nowadays. Nevertheless, AI is not a new topic. In 1955, McCarthy et al., “*the father of AI*” [47], uses the phrase AI in his proposal for the Dartmouth Summer Research Project on artificial intelligence [34].

According to a Gartner study in 2022¹, “*AI has gone through peaks and troughs over the past decades, but the last 10 years truly represent a breakthrough period in the annals of AI*” [10]. This evolution of AI would not have been possible without the convergence of five key factors [10]:

1. “*The rise of big data and parallel processing systems that can store and process data at massive scale, and cost-effectively*” [10].
2. “*Advancements in compute infrastructure, especially the emergence of powerful graphics processing units for complex computations*” [10].
3. “*The rise of new machine learning (ML) techniques - advancements in deep learning, reinforcement learning and the advent of transformer architectures that power large language models*” [10].
4. “*The emergence of cloud computing to experiment and operationalize AI faster with lower complexity*” [10].
5. “*The vibrant open-source ecosystem, which enabled mainstreaming of deep learning frameworks and resulted in an explosion of startups*” [10].

As the world grows more complex, materials and human resources must be used more efficiently [55]. Nevertheless, “*many people are under the false impression that the commercial goal of AI must be to save money by replacing human workers*” [55]. Moreover, and in contrast to this assumption, reaching new opportunities “*to identify trends at an early stage and to be able to respond to them using new business models*” [21] will often be more interesting than decreasing cost [55] [10].

¹ Building a Digital Future: Emergent AI Trends [10]

Additionally, and according to the Moravec’s paradox², the task of replacing human workers ranges from difficult to impossible because it is a tough challenge *“to endow computers with all the perceptions, reasoning, and action abilities that people exhibit”* [55]. Moreover, whereas AI captures a very wide range of technologies applied to decision-making problems in natural language processing, forecasting, analysis and optimization with a range of interpretations of data as things such as speech, video, robot movements, weather forecasting and human purchasing behavior [12, 55], *“AI does not include deeper human and animal abilities such as the creation of meaning, the connection with others and the ability to feel and think, except where aspects of meaning, connection, feeling and thinking can be encoded as data-in-data-out decision problems”* [12]. Thus, instead of replacing human workers and because intelligent people and intelligent computers have complementary abilities, *“people and computers can realize opportunities together that neither can realize alone”* [55].

Nowadays, whereas *“AI is a foundational catalyst for digital business”* [10], in reality most organizations struggle to scale their AI pilots into enterprise-wide production, which limits the ability to realize AI’s potential business value [23]. To be more precise, organizations *“are discovering ‘AI pilot paradox’, where launching pilots is deceptively easy but deploying them into production is notoriously challenging”* [23].

According to a KPMG study in 2021³, one reason for this situation is *“that the risks and challenges that AI poses, including codifying and reinforcing unfair biases and infringing on human rights such as privacy, are undeniable”* [31]. Thus, these issues are causing public concern and raising questions about the trustworthiness and regulation of AI applications [31]. *“If AI systems do not prove to be worthy of trust, their widespread acceptance and adoption will be hindered, and the potentially vast societal and economic benefits will not be fully realized”* [31].

Consolidated under the term *explainable artificial intelligence (XAI)* that represents processes and methods that allow human users to comprehend and trust the results and outputs created by machine learning algorithms [28], XAI is used to describe an ML model, its expected impact, and potential biases [49]. Thus, XAI helps characterize model accuracy, fairness, transparency, and outcomes in AI-powered decision-making [49]. Especially as AI becomes more advanced, whereas the whole analytics process is turned into what is commonly referred to as a “black box” that is impossible to interpret, humans have to be supported to comprehend and retrace how the machine learning algorithm came to a result.

“Within the last five years AI ethics has shifted from an academic concern to a matter for political as well as public debate” [16] and has also been recognized and discussed in the European Union [16]. Moreover, to support the implementation of their vision of *“ethical, secure, and cutting-edge AI made in Europe”*, the European Commission established the *High-Level Expert Group on Artificial Intelligence (AI HLEG)* [27] and the European Parliament decided and pronounced the first EU regulatory framework for AI (AI Act) in 2024 [20].

Particularly, because not even the engineers or data scientists who create an algorithm can understand or explain what exactly is happening inside such a black box or how the AI algorithm arrived at a specific result [49] [53], the AI HLEG highlights that the explainability as well as the trustworthiness⁴ of AI *“is crucial for an organization in building trust and confidence when putting AI models into production”* [49]. Thus, *“Explainable AI is one of the key requirements for implementing responsible AI”* [5].

Concluding the introduction of artificial intelligence, the importance of trustworthiness can be highlighted as an important factor for their widespread acceptance and adoption. In the following, current challenges in the context of AI-based big data analysis will be identified and precise motivation statements will be derived and presented next.

² *“It is comparatively easy to make computers exhibit adult level performance on intelligence tests or playing checkers, and difficult or impossible to give them the skills of a one-year-old when it comes to perception and mobility”* [35].

³ Trust in Artificial Intelligence: A five Country Study [31]

⁴ *“AI should be ‘lawful’, complying with all applicable laws and regulations; AI it should be ‘ethical’, ensuring adherence to ethical principles and values; and AI should be ‘robust’, both from a technical and social perspective, since, even with good intentions, AI systems can cause unintentional harm”* [27].

1.1 Motivation

“In a perfect world, free from budgets, every piece of data that is collectable would be collected, and every byte would be analyzed [...]” [2]. Big data analysis frameworks like, e.g., Hadoop, Spark, or R, have already found their way into mainstream application and have seen wide-spread deployment in scientific communities as well as in organizations across different industry fields [9]. Nevertheless, there is a gap between having access to these tools, which can leverage clusters of commodity computers for chugging through massive amounts of data, and doing something useful with it, not only a so-called data scientist might care about [41]. *“Only an organization’s technical ability for big data analysis does not automatically result in corresponding availability of competent human resources for utilizing this infrastructure”* [9]. Moreover and as already introduced, by integrating AI-supported big data exploration, humans are challenged to comprehend and retrace how the algorithm came to a specific result [49] [53]. *“If AI systems do not prove to be worthy of trust, their widespread acceptance and adoption will be hindered, and the potentially vast societal and economic benefits will not be fully realized”* [31]. Without the strong technical background of data scientists, who are capable of both extracting meaningful value from collected data and managing the entire life cycle of data, including supporting data infrastructures [26] [32], designing, establishing, operating, and customizing of modern and trustworthy AI-based big data analysis infrastructures will be difficult to achieve (*Motivation Statement 1*).

Traditional *“big data analysis is based on different perspectives and intentions”* [9]. Thus, business domain users of big data analysis applications have to perceive and understand what big data analysis results stand for. This has to be quite often achieved without an understanding of big data analysis itself [9]. Moreover, Kaufmann [30] describes big data management as *“the process of optimally controlling flows of large-volume, high-velocity, heterogeneous, and uncertain data”*. Usually, big data is approached with a technological focus, but a key question for businesses is how big data can be effectively used to create value [9]. In order to satisfy these motivations, a theoretical reference model (IVIS4BigData), that is based on Kaufmann’s Reference Model for Big Data Management (c.f. Figure 5) has been developed, evaluated, and implemented in different real-world application scenarios within earlier related work [9]. Nevertheless, whereas on the one hand, IVIS4BigData covers the whole workflow of big data analysis from raw data sources into views that finally enable the different user stereotypes participating in the process to gain insight, on the other hand, IVIS4BigData lacks trustworthy AI-supported data exploration within the whole big data analysis process. However, AI-support is a necessary requirement for modern (big data) analysis applications (*Motivation Statement 2*).

To extend IVIS4BigData, the integration of (trustworthy) AI-supported data exploration has already been identified as a remaining challenge within the authors dissertation project [9]. Moreover, although IVIS4BigData has been successfully developed, evaluated, and implemented in different proof-of-concept real-world application scenarios (c.f. [8] [40]), there is still a need to deploy and demonstrate their applicability and impact in a productive industrial application solution (*Motivation Statement 3*).

An industrial application domain highlighting the necessity of trustworthy AI-supported data exploration in a real-world big data analysis application scenario is addressed in the DARIA⁵ project of the STRABAG Group as an exemplary representative of a large industrial multinational enterprise (exemplary industrial use case scenario). Currently, the economic risk analysis of construction projects is largely based on so-called domain expert knowledge, an experience- and intuition-based analysis of the risks as well as on various statistical and quantitative analyses of projects (applying manual methods) that have already been completed. Even though experience-based processes can be organized and formalized in different ways like, e.g., brainstorming, checklist analysis, literature and document review, workshops, cause and effect diagrams, or SWOT⁶-analysis [13], an IT-based automation support is needed. Unfortunately, the main problem is that construction projects of, e.g., the STRABAG group *“are characterized by carrying a high level of uncertainty and complexity”* [13]. It is very likely that this holds true for other industrial enterprises.

⁵ Data-driven Risk Analysis in the Construction Industry [48]

⁶ Strength, Weakness, Opportunity, and Threat

In addition to the previous outlined exemplary industrial use case scenario, also further research projects at the University of Hagen, Faculty of Mathematics and Computer Science, like, e.g., ORIKA⁷ or AI4H3⁸ (exemplary medical use case scenarios), which are based on IVIS4BigData, require trustworthy artificial intelligence functionalities within their big data analysis. As an example, clinical AI interpretation is a central element within the AI4H3 KlinGard open software platform.

Although modern hospital information systems are supporting patient screening and prevention as well as diagnosis, prognosis, or treatment pathways and are well-established in Europe, their primary focus is mainly storing and retrieving patient records and data [17]. Therefore, AI4H3 proposes deploying assistive AI techniques for automatically supporting the processing of patients' medical records and medical data for, e.g., automatically supporting the assistive recognizing of salient medical features in several types of medical data and multimedia content and representing them in a machine-readable and at the same time semantic way [17].

Nevertheless, especially in the healthcare sector, even traditional software applications without AI functionalities are under a strong regulation⁹ [19]. If a software participates in conducting a medical task (*medical device software (MDSW)*) and unintended use of this software is bound to high risks, in the European Union for example, this software falls under *medical devices regulation (MDR)*. In the United States, such software requires an FDA¹⁰ approval. Thus, both AI research projects ORIKA and AI4H3 that are assigned to this sector are potentially under special focus of regulation. Even though a general regulation in this sector is not yet very pronounced, trustworthiness of AI is a crucial factor as well.

Concluding the outlined challenges of the application scenarios in science and industry, the importance of trustworthy AI-supported data exploration in big data analysis applications has been highlighted. Unfortunately, IVIS4BigData does not meet these new requirements and must therefore be adapted to the new circumstances in terms of trustworthy artificial intelligence.

1.2 Problem Statements and Research Questions

In summary, the benefit and promise of AI for science, industry, and society are undeniable. On the one hand, AI is helping people make better predictions and informed decisions, enabling innovation and productivity gains [31]. On the other hand, the risks and challenges that AI poses, including codifying and reinforcing unfair biases and infringing on human rights such as privacy, are also undeniable [31].

Therefore and based on the motivation statements 1-3; *modeling of trustworthy artificial intelligence, trustworthy AI-support as a necessary requirement for modern (big data) analysis applications* as well as *the need to deploy and demonstrate their applicability and impact in productive industrial application solutions*, the contributions summarized in this research focuses on **Trustworthy AI supporting Big Data Analysis**. Encouraged by the importance that business domain users need opportunities to understand, to comprehend, and therefore to trust and to deploy AI applications successfully into production, this research will be guided by the following problem descriptions, problem statements, and research questions:

Problem Description 1: During the process of implementing new and innovative software applications, the challenge of acceptance arises. In order to enable human users to have confidence in the technology's development and its application, modeling trustworthiness in AI applications is essential to deploy and use AI systems. Nowadays, however, many AI applications are often focusing on the technological perspective of solving the problem being addressed. Especially when these applications support human users in their decision-making process, the internal data processing of an AI application needs to be as transparent as possible, the decisions made by the AI application need to be explained as appropriate as possible, and the overall purpose as well as the capabilities of the AI application must be openly communicated to those directly and indirectly affected.

⁷ Modeling Metagenomics, Microbiota, and Inflammation Biomarkers to Deliver a Personalized Blueprint of Chronic Inflammation and Care Delivery in Health-to-Disease Transition Management [18]

⁸ Artificial Intelligence for Hospitals, Healthcare and Humanity [17]

⁹ e.g., AI Act in the European Union [20]

¹⁰ US Food and Drug Administration

⇒ **Problem Statement 1:** In this way, the lack of modeling of trustworthiness in context of artificial intelligence in big data analysis application scenarios is identified as the first problem statement.

⇒ **Research Question 1:** To address these challenges within the first problem description and the resulting problem statement, the following research question has been defined: **How can trustworthiness of artificial intelligence in big data analysis be modeled, implemented, and evaluated?**

Problem Description 2: AI is often associated with the use of machine learning algorithms for data analysis. Nevertheless, and from closer look on big data analysis application, more process steps are involved in the analysis process of big data (c.f. IVIS4BigData [9]). Moreover, and to enable end users to use, change, and enhance their applications and build new ones without having to become professional programmers or data scientists, also human-computer interaction including end user empowerment are important components in these applications. Thus, the relevance of trustworthy AI cannot be limited only to the big data analysis process step itself. For example, also the selection of suitable big data analysis algorithms or methods, the selection of necessary data sources or the recommendation of meaningful visualizations, can be possible AI use cases.

⇒ **Problem Statement 2:** In this way, the lack of modeling of the integration of trustworthy artificial intelligence into the whole big data analysis process to generate added value is identified as the second problem statement.

⇒ **Research Question 2:** To address these challenges within the second problem description and the resulting problem statement, the following research question has been defined: **Where and how can trustworthy artificial intelligence be integrated into the whole big data analysis process?**

Problem Description 3: Although AI has become extremely popular nowadays, in reality most organizations struggle to scale their AI pilots into enterprise-wide production. Also, traditional and non-highly regulated industries like, e.g., the construction industry faces the challenge of acceptance and adoption of AI applications. One reason is that AI is often used to solve problems that are not AI compatible. According to Moravec’s paradox, AI should be used in decision-making application scenarios, such as pattern matching, anomaly detection, or other application scenarios where trustworthy AI can help a human user within their decision-making process, instead of using AI for applications that solve sensorimotor application scenarios, such as perception and mobility. Nevertheless, even if trustworthy AI is used for the right application scenarios, the question of the best possible or most suitable application remains. Instead of replacing human workers and because intelligent people and intelligent computers have complementary abilities, *“people and computers can realize opportunities together that neither can realize alone”* [55].

⇒ **Problem Statement 3:** In this way, the elaboration of how trustworthy artificial intelligence can be integrated in real-world big data analysis application scenarios is identified as the third problem statement.

⇒ **Research Question 3:** To address these challenges within the third problem description and the resulting problem statement, the following research question has been defined: **How can trustworthy artificial intelligence be integrated in real-world productive big data analysis application scenarios?**

1.3 Methodology and Approach

Research can be defined as *“systematic, intensive study directed toward fuller scientific knowledge of the subjected study”* [6]. This process *“involves understanding the research domains, asking meaningful research questions, and applying valid research methodologies to address these questions”* [36].

Thus, a suitable research methodology that *“consists of the combination of the processes, methods, and tools that are used in conducting research in a research domain”* [36] has to be utilized. With focus on this *information systems (IS)* research, the research methodology framework for IS, proposed by Nunamaker et al. [36] and illustrated in Figure 1, has been selected as a methodological framework.

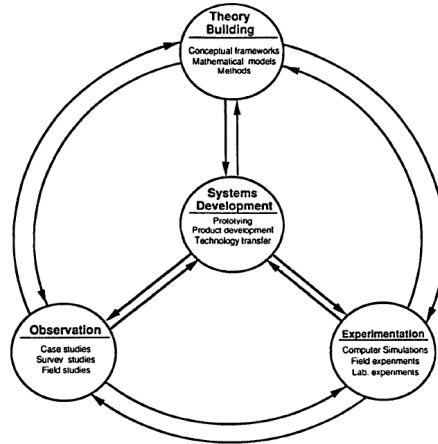


Fig. 1. A Multimethodological Approach to IS Research [36]

This research methodology framework, consisting of four research strategies *observation*, *theory building*, *systems development*, and *experimentation*, provides a systematic orientation that is necessary “if IS research is to keep pace with technological innovation and organizational acceptance” [36]. These research strategies, which include methods, do not have a fixed execution order. Although each strategy is connected directly to each other, system development can be considered as a central element within the IS methodology framework, which is supported by a surrounding strategy.

Theory building includes the development of innovative ideas, concepts, or the construction of conceptual frameworks [36]. The resulting theories are often used “to suggest research hypotheses, guide the design of experiments, and conduct systematic observations” [36]. Experimentation addresses the gap between theory building and observation including methods, like experiments and simulations [36]. Observations include unobtrusive research methods, like studies and surveys, to help researchers to formulate specific hypotheses or to generate a universal feeling for the challenges in a research domain [36]. Finally, systems development consists of several system design methods, starting from prototyping up to technology transfer [36]. In this way, this methodology framework provides a general structure for the presented research.

In order to ensure a structured approach, the defined research questions RQ1 - RQ3 are addressed by planning and implementing specific R&D projects, where each project applies the introduced Nunamaker [36] methodology framework. Figure 2 outlines this structured research approach.

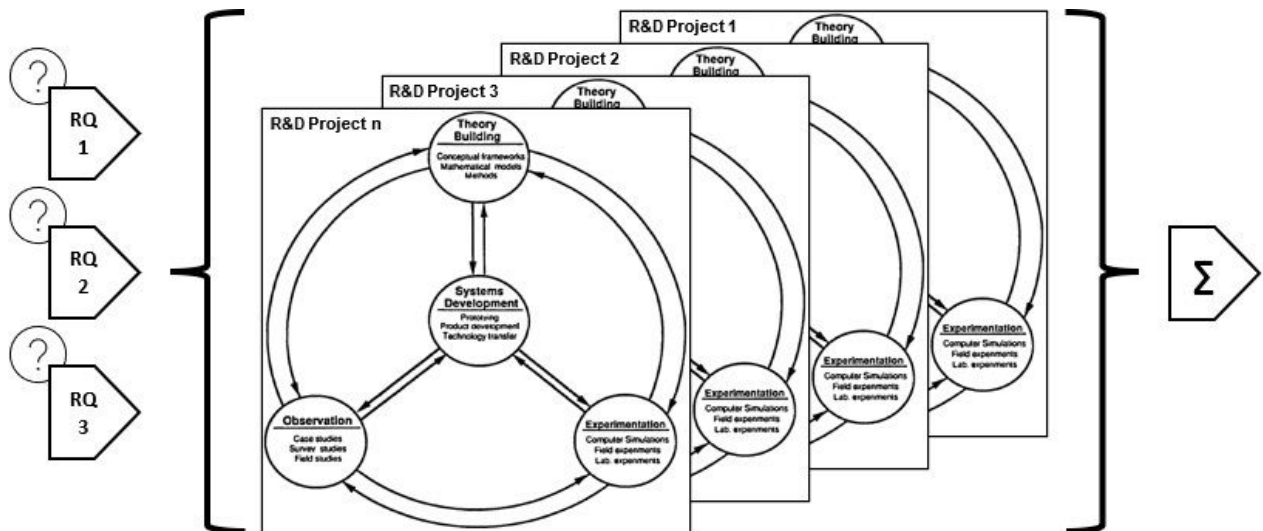


Fig. 2. Outline of the R&D Projects in Context of the Research Methodology

Whereas this research with its underlying three research questions motivates the overall goals of the research program by means of targeted R&D projects, each of the R&D projects addresses partial aspects of the research questions using the same methodology to ensure a comparability and interoperability between the research projects. Finally, the contributions will be summarized in the author's habilitation thesis.

The conceptual methodological schema of the overall research program, to be planned and implemented by certain R&D projects, represents the first step in the definition of this structured research approach. As important second step, the R&D projects have to be defined and their outcomes have to be assigned to the research questions RQ1 - RQ3. Figure 3 illustrates the conceptual methodological schema of the overall research program including an exemplary breakdown and assignment of research goals in future work R&D projects.

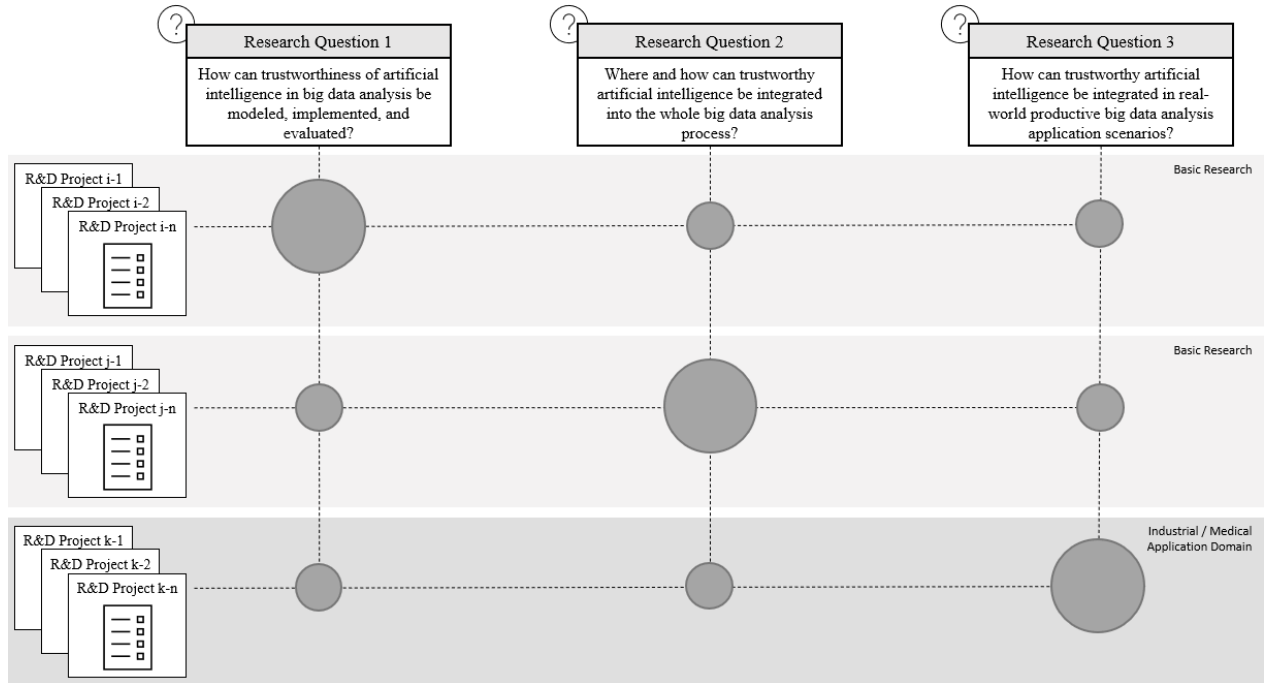


Fig. 3. Exemplary Conceptual Methodological Schema and Assignment of Research Goals in Future Work R&D Projects

As outlined in this conceptual methodological schema illustration, the outcomes of each R&D project are represented as circles. Since each R&D project potentially contributes to more than one research question, the surface area of the circles indicates the contribution priority. The larger the surface area, the larger the contribution to the research question. With the aid of this illustration, the focus of each R&D project can be optimally defined to ensure a large as possible coverage of the research questions.

After the introduction of artificial intelligence into big data analysis processes, the importance of supporting trustworthiness as an important factor for their widespread acceptance and adoption has been highlighted. Afterwards, relevant challenges and motivation statements in the context of AI-based big data analysis have been identified before the introduction of an exemplary industrial application domain highlighting the necessity of AI-supported data exploration in different real-world big data analysis application scenarios has been outlined. Moreover, precise descriptions of relevant problem statements and corresponding research questions as a basis for this research approach have been defined. In the remainder of this paper, an initial review of the relevant state of the art in science and technology as well as related work will be presented next to identify remaining challenges. Based on these results, a conceptual reference model for trustworthiness in AI-based big data analysis will be introduced and discussed.

2 State of the Art in Science and Technology

Related to the overall goal of this research (*Trustworthiness in AI-based Big Data Analysis*) as well as to enable a modeling of trustworthiness in AI-based big data analysis (*RQ1*), the initial review of the relevant state of the art in science and technology is focusing on the definition of intelligence, artificial intelligence, trust, and trustworthiness first. Afterwards and to integrate trustworthy artificial intelligence into the whole big data analysis process (*RQ2*) as well as to integrate trustworthy artificial intelligence in real-world productive big data analysis application scenarios (*RQ3*), relevant related work will be outlined and discussed.

As already addressed in the introduction of this research report, artificial intelligence is not a new topic. In 1955, McCarthy et al., “*the father of AI*” [47], uses the phrase AI in his proposal for the Dartmouth Summer Research Project on Artificial Intelligence [34]. Nevertheless, before artificial intelligence can be introduced further, the general concept of intelligence needs to be discussed first.

In 1904, the English psychologist Charles Spearman described intelligence as a “*general capacity of the individual to think rationally and problem solving*” [46]. Moreover, Spearman structured intelligence in two factors; the general ability known as *general factor* (g-factor) and specific abilities known as specific factors (s-factor) [46]. Whereas the g-factor, which is not affected by learning or experience, represents a general capacity that help an individual to perform all the mental or intellectual activity, the nature of the s-factors of an individual are changeable and are affected by learning or experience [46].

In 1924, the American psychologist Louis Leon Thurstone formulated a model of intelligence centered on primary mental abilities [50] [52]. Thurstone opposed the notion of a singular general intelligence (g-factor) [51]. By the aid of fifty-six tests, selected according to a provisional classification of cognitive factors that were applied to 240 volunteers, seven independent group factors of intelligence¹¹ were identified. In contrast to Spearman, Thurstone emphasizes that all seven abilities can grow at different rates in an individual whereas the growth of one ability affects and is affected by others [51]. Thus, the total intelligence of an individual comes from the growth of many abilities [51].

Even though newer approaches from cognitive psychology led to an expansion of the concept of intelligence and formulated the assumption of several forms of intelligence (c.f. Guilford’s Structure of Intellect [24] or Gardener’s Theory of Multiple Intelligences [22]), intelligence is quite diverse and its expression can only be determined through natural observation and assessment of an individual in different real-world scenarios. This is important when defining and also evaluating artificial intelligence.

Rationally, the International Organisation for Standardisation (ISO) [29] describes AI as a form of intelligence displayed by machines other than natural intelligence which humans and animals display [29]. Since the early 1950s, “*when Alan Turing first posed the question of whether machines can think*” [37] [54], the artificial intelligence technical landscape has evolved significantly [37]. Initially defined as “*the science and engineering of making intelligent machines*” [34], “*AI has evolved from symbolic AI where humans built logic-based systems, through the AI ‘winter’ of the 1970s to the chess-playing computer Deep Blue in the 1990s*” [37].

To be more precise, “*research has historically distinguished symbolic AI from statistical AI*” [37]. Symbolic AI uses logical representations to deduce a conclusion from a set of constraints and requires that researchers build detailed and human-understandable decision structures to translate real-world complexity and help machines arrive at human-like decisions [37]. “*Symbolic AI is still in widespread use, e.g., for optimization and planning tools*” [37]. On the other hand, statistical AI, “*whereby machines induce a trend from a set of patterns, has seen increasing uptake recently*” [37].

One example for statistical AI is machine learning which represents a set of techniques to allow machines to learn in an automated manner through patterns and inferences rather than through explicit instructions from a human [37]. “*ML approaches often teach machines to reach an outcome by showing them many examples of correct outcomes*” [37].

After the general introduction of the concept of intelligence, a brief description of artificial intelligence has been discussed. Nevertheless, and as outlined within the introduction, “*if AI systems do not prove to be worthy of trust, their widespread acceptance and adoption will be hindered, and the potentially vast societal and economic benefits will not be fully realized*” [31]. In this way, and following the overall goal of this research, trust as well as trustworthiness will be introduced and discussed next.

¹¹ Space, perceptual speed, numerical ability, memory, reasoning, word fluency, and verbal relations.

Although trust has been shown as a crucial factor for the adoption of new technologies, “*surprisingly, trust literature offers very little guidance for systematically integrating the vast amount of behavioral trust results into the development of computing systems*” [45]. Rationally, trust is defined as “*firm belief in the reliability, truth, or ability of someone or something; confidence or faith in a person or thing, or in an attribute of a person*” [38].

Nevertheless, trust is much more ambivalent [43]. This becomes clear when the term is approached philosophically. David Easton, a Canadian American political scientist, highlights “*that trust is based on experience with the outputs of decision-makers*” [15]. “*In contrast to specific support, however, it does not refer to the daily outputs of concrete decision-makers, but to ongoing positive experiences with outputs of various decision-makers over a longer period of time*” [15]. In this way, Easton defines trust based on the *reliability* and *competence* as well as on the output of an individual or system instead of an individual or system itself. This becomes even clearer when it is combined with Annette Baier’s statements. Baier, a New Zealand philosopher, emphasizes that “*trust is much easier to maintain than it is to get started and is never hard to destroy*” [4]. Moreover, “*trust is a fragile plant, which may not endure inspection of its roots, even when they were, before the inspection, quite healthy*” [4].

Trust is often discussed when it becomes fragile, when it is no longer perceived as a given background, but when it is questioned [43]. Therefore, trust can be defined based on the *honesty* of an individual and plays a significant role where there is no security, no certainty [43]. In this way, trust is not only a crucial factor for the adoption of new technologies. Especially in the context of AI, where the analytics process is turned into what is commonly referred to as a “black box” that is impossible to interpret and therefore humans are challenged to comprehend and retrace how the algorithm came to a result.

Summarizing the discussion about trust, between individuals, trust is on the one hand based in terms of their *competence* and on the other hand based in terms of their *honesty* [43]. Therefore, trust and trustworthiness have at least both a knowledge component and a moral component [25]. With regard to initial rational definition of trust (reliability, truth, or ability), ability and truth can be replaced by competence and honesty, whereas reliability remains as an important third attribute according to the interpretation of Easton’s argumentation. Figure 4 illustrates the derived dimensions of trust in the interaction between individuals.

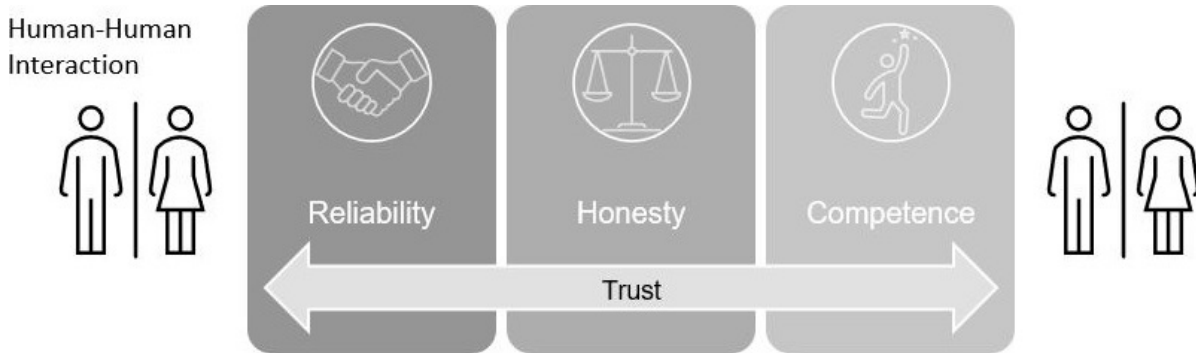


Fig. 4. Dimensions of Trust in the Interaction Between Individuals

Even if these dimensions of trust are suitable for individuals, they are inappropriate for describing a system. To enable a modeling of trustworthiness in AI-based big data analysis (*RQ1*), corresponding dimensions that are able to define trustworthiness based on the output of an individual or system instead of an individual or system itself [15] need to be defined. In this way, the definition of corresponding dimensions of trust can be identified as *Remaining Challenge 1*.

Related to the overall goal of this research (*Trustworthiness in AI-based Big Data Analysis*) as well as to enable a modeling of trustworthiness in AI-based big data analysis (*RQ1*), the definition of intelligence, artificial intelligence, trust, and trustworthiness have been introduced and discussed first. Afterwards and

to integrate trustworthy artificial intelligence into the whole big data analysis process (*RQ2*) as well as to integrate trustworthy artificial intelligence in real-world productive big data analysis application scenarios (*RQ3*), relevant related work will be outlined and discussed next.

Kaufmann [30] describes *big data management (BDM)* as “*the process of optimally controlling flows of large-volume, high-velocity, heterogeneous, and uncertain data*”. Usually, big data is approached with a technological focus, but a key question for businesses is how big data can be effectively used to create value [9].

Therefore, in this model and as illustrated in Figure 5, the classical technological aspects of big data (c.f. Singh and Reddy [44]), namely *data integration* (i.e. Hadoop [3] clusters, NoSQL databases, and other database management systems) and *data analytics* (i.e. statistical, machine learning, and data mining tools and techniques) are complemented by three additional layers, called *data interaction*, *data effectuation*, and *data intelligence* to improve the effective benefit of big data technologies [7].

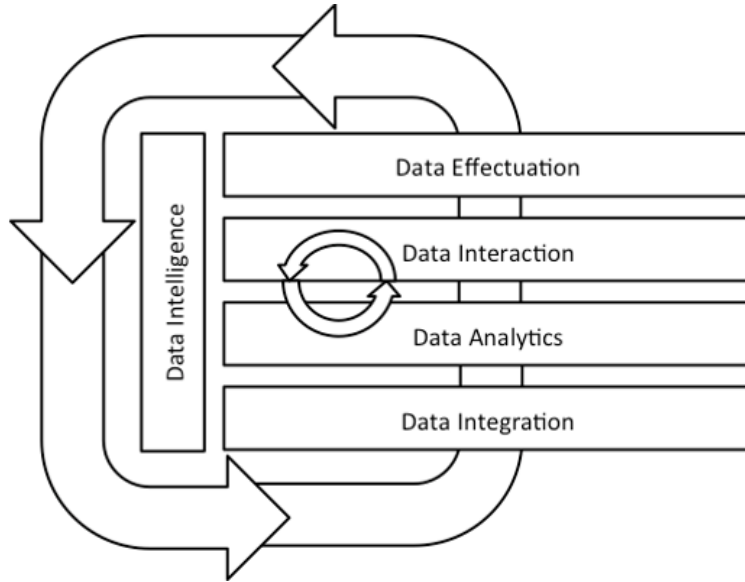


Fig. 5. Reference Model for Big Data Management [30]

On top of the two basic layers which are focusing on data technology, there is a layer called data interaction to illustrate the importance of human-computer interaction in the process for big data management which is defined as a key aspect in this model [9]. In this step, human decision makers are getting in touch with analysis results to view, manipulate, correct, and communicate them. This bi-directional interaction between technology and its users is symbolized by the small feedback loop [9].

Nevertheless, interacting with analysis results is not sufficient to create actual value from data. For the optimization of business objectives, according to Davenport [14], it is necessary that the effects of data analysis are integrated with products and services of an organization. Therefore, another layer called data effectuation is defined in which value creation from data is addressed. Finally, the model describes a cross-sectional function of knowledge-based processes and technologies which support the big data management life cycle [9]. Value creation from data depends highly on emergent knowledge processes in the organization [33] [39]. The layer called data intelligence ought to ensure that knowledge and skills can be acquired and properly managed in the context of big data management. Kaufmann [30] proposes to apply knowledge management and knowledge engineering techniques to all layers of the big data management from *integration* to *effectuation*, for example, by archiving and semantically annotating analysis results, by optimally communicating data-based insights, or by actively managing data science know-how [7]. In this way, i.e., by actively managing the insights created by big data analytics and by iteratively effectuating data and integrating these effects into the knowledge base, an emergent knowledge process of higher order between the organization and its environment is created which is symbolized by the outer loop [7].

Nevertheless, even if Kaufmann’s BDM Reference Model describes an emergent knowledge process of higher order between a big data analysis system and its environment, it lacks trustworthy AI functionalities. To integrate trustworthy artificial intelligence into the whole big data analysis process (*RQ2*), this big data management reference model needs to be refined to a trustworthy AI-based big data management reference model. In this way, the integration of the technical aspects of trustworthy artificial intelligence into this big data management process can be identified as *Remaining Challenge 2*.

Additionally and with focus on the third research question to integrate trustworthy artificial intelligence in real-world productive big data analysis application scenarios (*RQ2*), AI Group of Experts at the OECD (AIGO) high-level structure of a generic AI system also represents a relevant reference model for this research. Figure 6 illustrates the high-level structure of a generic AI system (also referred to as *intelligent agent*) [37].

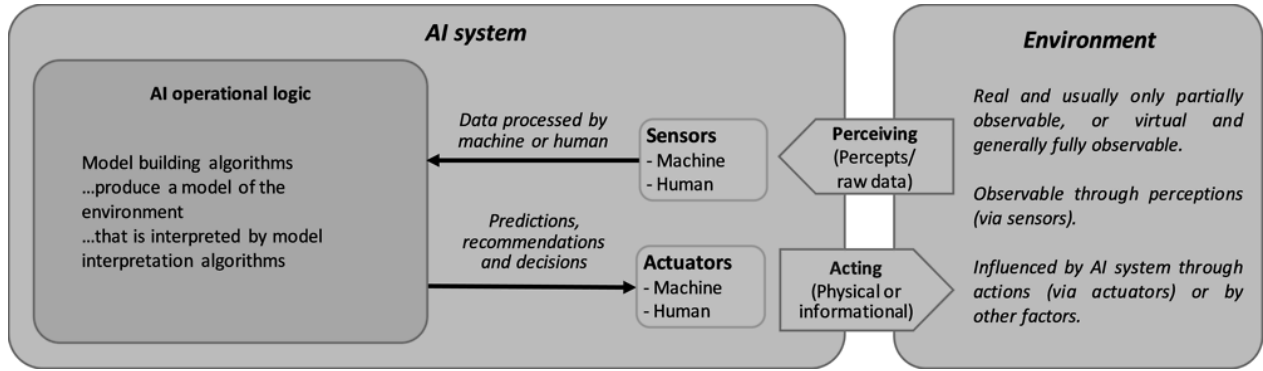


Fig. 6. AIGO’s High-Level Conceptual View of an AI System [37]

In 2018, AIGO set up a subgroup to develop a description of an AI system “that aims to be understandable, technically accurate, technology-neutral, and applicable to short- and long-term time horizons” [37]. AIGO’s high-level conceptual view of an AI system “aims to be understandable, technically accurate, technology-neutral and applicable to short- and long-term time horizons. It is broad enough to encompass many of the definitions of AI commonly used by the scientific, business and policy communities” [37]

According to this concept, an AI system consists of three main elements: Sensors, operational logic, and actuators [37]. Whereas sensors collect raw data from the environment and actuators act to change the state of the environment, “the key power of an AI system resides in its operational logic” [37]. For a given set of objectives and based on input data from sensors, the operational logic provides output for the actuators, which take the form of recommendations, predictions or decisions that can influence the state of the environment [37].

AIGO’s high-level structure of a generic AI system can close the gap between the modeling of a trustworthy AI-based big data analysis process and its interaction between with its environment within a real-world productive big data analysis application scenario. In this way, the combination of the refined BDM Reference Model (*Remaining Challenge 2*) with the AIGO’s high-level structure of a generic AI system can be identified can be identified as *Remaining Challenge 3*.

Concluding the initial review of the relevant state of the art in science and technology and related to the overall goal of this research (*Trustworthiness in AI-based Big Data Analysis*) as well as to enable a modeling of trustworthiness in AI-based big data analysis (*RQ1*), the definition of intelligence, artificial intelligence, trust, and trustworthiness have been introduced and discussed first. Afterwards and to integrate trustworthy artificial intelligence into the whole big data analysis process (*RQ2*) as well as to integrate trustworthy artificial intelligence in real-world productive big data analysis application scenarios (*RQ3*), relevant related work has been introduced and discussed. In the following, the conceptual modeling and design of trustworthiness as well as the integration of artificial intelligence into the whole big data analysis process will be outlined.

3 Conceptual Modeling and Design

Following the general structure of this research, the conceptual modeling of trustworthiness in AI-based big data analysis (*RQ1*) will be outlined first. Afterwards, the integration of trustworthy artificial intelligence into the whole big data analysis process (*RQ2*) as well as the integration of trustworthy artificial intelligence in real-world productive big data analysis application scenarios (*RQ3*) will be modeled and discussed.

Revisiting the state of the art of trust and trustworthiness, both have at least a knowledge component and a moral component [25]. With regard to the initial rational definition of trust (reliability, truth, or ability) [38], ability and truth can be replaced by competence and honesty, whereas reliability remains as important third attribute according to the interpretation of Easton's [15] and Baier's [4] argumentation. Nevertheless, this definition (*reliability, honesty, and competence*) only represents the dimensions of trust in the interaction between individuals (c.f. 4).

Even if these dimensions of trust are suitable for individuals, they are inappropriate for describing a system. Therefore, corresponding dimensions that are able to define trustworthiness based on the output of an individual or system instead of an individual or system itself [15] need to be defined (*Remaining Challenge 1*). With focus on the first trust dimension *reliability*, which describes the quality of individuals of performing consistently, *reproducibility* can represent this dimension in context of a system by describing the ability “of obtaining consistent computational results using the same input data, computational steps, methods, code, and conditions of analysis” [42]. Moving forward to the next dimension *honesty*, which can be defined as adherence to the facts between individuals, *validity*, which describes the degree to which data or the results is correct, can represent this second dimension of trust in context of a system. Finally, *competence*, which basically describes the ability of a person to do something well, can be represented by the term *capability*, which describes the general potential of a system to accomplish something. Figure 7 illustrates the dimensions of trust in the interaction between individuals as well as the dimensions of trustworthiness of a system in the interaction between individuals and systems.

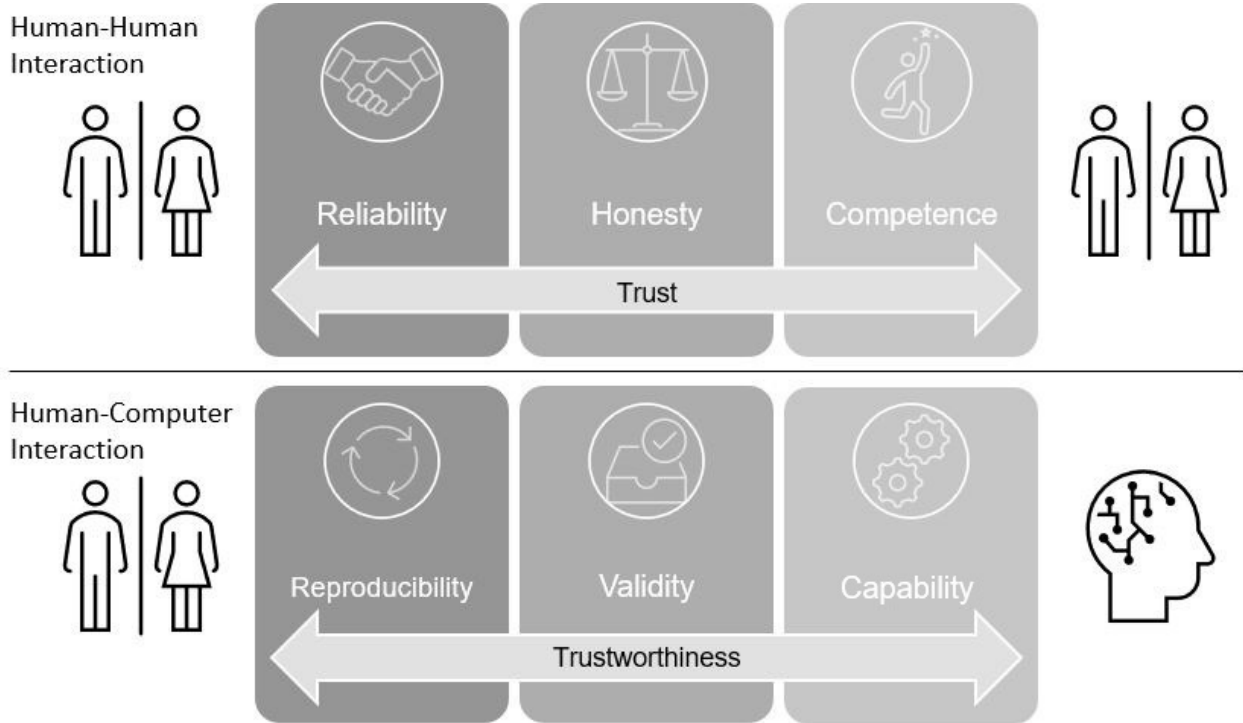


Fig. 7. Dimensions of Trust and Trustworthiness in the Interaction Between Individuals as Well as Between Individuals and Systems

With focus on AI HLEG’s ethics guidelines for trustworthy artificial intelligence [27], which lists seven key requirements¹² that AI systems should meet in order to be trustworthy [27], it is important to emphasize that these requirements are applicable to different stakeholders (*developers, end users, as well as broader society*) partaking in AI systems’ life cycle [27].

Motivated by the importance that business domain user need opportunities to understand, to comprehend, and therefore to trust and to deploy AI applications successfully into production (c.f. Section 1.2), this research focuses on AI systems in productive research or industrial application solutions. In this way, relevant stakeholders of an AI-system in this limited scope are developers as well as specialized business domain users. Moreover, due to the resulting loss of the third user group *broader society*, also ethical aspects are less important in this research. Nevertheless and in alignment with the AI HLEG’s guideline, all dimensions of trustworthiness of a system in the interaction between individuals and systems “*are of equal importance, support each other, and should be implemented and evaluated throughout the AI system’s life cycle*” [27].

After the definition of trustworthiness in the interaction between individuals and systems addressing the modeling of trustworthiness in AI-based big data analysis (*RQ1*), the integration of trustworthy artificial intelligence into the whole big data analysis process (*RQ2*) as well as the integration of trustworthy artificial intelligence in real-world productive big data analysis application scenarios (*RQ3*) remain open. Moving forward to the second research question, Kaufmann’s BDM Reference Model (c.f. Figure 5) as well as Easton’s definition that trust is based on the output of an individual or system instead of an individual or system itself [15] can pave the way of modeling trustworthiness in AI-based big data analysis.

As outlined within the state of the art, Kaufmann’s BDM Reference Model describes an emergent knowledge process of higher order between a big data analysis system and its environment. Nevertheless, it lacks trustworthy AI functionalities. In this way, although this reference model paves the way of modeling trustworthiness in AI-based big data analysis, it needs to be refined to a trustworthy AI-based big data management reference model (*Remaining Challenge 2*).

Kaufmann’s reference model contains two different types of layers. Four technical layers on the right side that improve the effective benefit of big data technologies to represent the technical analysis process as well as one cross-functional layer “*to ensure that knowledge and skills can be acquired and properly managed in the context of big data management*” [30]. In this way, the integration of the technical aspects of AI into the whole big data analysis process are located on the right side, whereas the integration of the aspects regarding to trustworthiness are located on the left side within the *data intelligence* layer. Figure 8 illustrates the integration of the technical aspects of trustworthy artificial intelligence into Kaufmann’s BDM Reference Model.

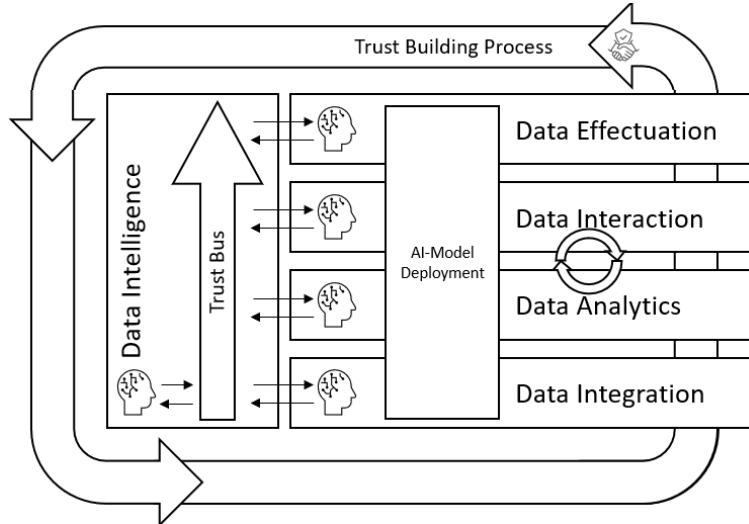


Fig. 8. Integration of Trustworthy Artificial Intelligence Into Kaufmann’s BDM Reference Model
¹² Human Agency and Oversight, Technical Robustness and Safety, Privacy and Data Governance, Transparency, Diversity, non-Discrimination and Fairness, Societal and Environmental Wellbeing, and Accountability.

With focus on the right side, technical aspects of artificial intelligence into the whole big data analysis process are located in all *data integration*, *data analytics*, *data interaction*, and *data effectuation* layers. The second *data analytics* layer represents the most important layer in context of artificial intelligence at first glance. Nevertheless, artificial intelligence also plays an important role within the other three layers.

Starting with *data integration*, artificial intelligence can be utilized within the combination process of data from different distributed and heterogeneous data sources into a single platform with consistent access [30]. To be more precise, AI-based data integration can improve a wide range of data integration tasks such as the mapping of different data schemas and standards, the combination of batch and real-time data, the identification and correction of data issues as well as the automatic recognition of master data domains, attributes, and hierarchies. In this way, AI can support the data integration process *“that data can become analyzable”* [30].

Data analytics includes *“the transformation of raw data into useable information”* [30]. This layer represents the most important layer in context of AI because the application of data science that is defined as *“the extraction of actionable knowledge directly from data through a process of discovery, or hypothesis formulation and hypothesis testing”* [11] takes place in this layer. Whereas all AI-related layers can implement symbolic AI approaches, which use logical representations to deduce a conclusion from a set of constraints and require that researchers build detailed and human-understandable decision structures to translate real-world complexity and help machines arrive at human-like decisions [37], artificial intelligence within the *data analytics* implements primarily statistical AI approaches that allows machines to learn in an automated manner through patterns and inferences rather than through explicit instructions from a human [37]. In this way, XAI that is used to describe an ML model, its expected impact, and potential biases [49], is especially an important factor in this layer because humans have to be supported to comprehend and retrace how the machine learning algorithm came to a result.

Data interaction consists of mutual interferences of users and data analysis results that generate individual and organizational knowledge, *“which is important to note that data analysis results are in fact nothing but more data unless users interact with them”* [30]. To be more precise and according to the AI4VIS¹³ study, visualizations can be created, stored, shared, and (re-)used with artificial intelligence techniques [56]. In this way, artificial intelligence can be utilized to support the generation and the enhancement of visualizations as well as to support the resulting human analysis process of the generated visualizations within the interaction between humans and the big data analysis results. *“In this step, human decision makers are getting in touch with analysis results to view, manipulate, correct, and communicate them. Conversely, data are getting in touch with the users of the information system, creating a bi-directional interaction between technology and its users, as symbolized by the small feedback loop”* [9].

Even if the fourth *data effectuation* layer focuses *“on the utilization of the data analysis results to create value in products, services, and operations of the organization”* [30], the interaction between individuals and systems also plays an important role here. Therefore, artificial intelligence can also be utilized in this layer.

Finally and with regards to AIGO’s system description of an AI system life cycle [37], *“the centrality of data and of models that rely on data for their training and evaluation distinguishes the life cycle of many AI systems from that of more general system development”* [37]. In this way, a cross-functional *model deployment* layer is defined next to the small feedback loop that highlights the iterative process of planning and designing, building and interpretation, verification and validation, as well as the deployment of AI models into enterprise-wide production [37].

With focus on the left side, *data intelligence* refers to the ability of the organization to acquire knowledge and skills in the context of big data management [30]. *“This can be understood as knowledge management and knowledge engineering applied for all steps of the data life cycle”* [30]. Moreover, *data intelligence* is a *“knowledge-driven cross-platform function that ensures that knowledge assets are optimally deployed, distributed, and utilized over all four layers of big data management”* [30]. In this way, artificial intelligence can also be utilized in this layer to support the overall knowledge management and knowledge engineering process.

¹³ AI4VIS: Survey on Artificial Intelligence Approaches for Data Visualization [56]

Revisiting Easton’s definition that trust is based on the output of an individual or system instead of an individual or system itself [15] in combination with the integration of artificial intelligence within all layers, each layer with AI-support needs an individual trust-component based on its output within the *data intelligence* layer. Moreover, the overall trustworthiness of a system depends on the trustworthiness of the outcome of the whole process. Thus, the individual trustworthiness of the AI-supported layers needs to be integrated into an overall *trust bus*. In this bus, the dimensions of trustworthiness in the interaction between individuals and AI components (c.f. Figure 7) within the certain layers as well as information about past interactions between individuals and the overall AI system within “*the emergent knowledge process of higher order*” [30], which is symbolized by the outer loop of the conceptual reference model for trustworthiness in AI-based big data analysis, need to be stored to gauge trustworthiness of the overall AI-based big data analysis to enable a trustworthy AI [1]. Nevertheless, trustworthiness itself is located at the interface between the system and individuals according to Easton, or between the AI system and the environment according to AIGO’s high-level structure of a generic AI system. In this way, the overall *trust building process* between individuals and the system itself is located in the outer loop.

Moving forward to the third research question, AIGO’s high-level structure of a generic AI system (c.f. Figure 6) can close the gap between the modeling of a trustworthy AI-based big data analysis process and its interaction with its environment within a real-world productive big data analysis application scenario (*Remaining Challenge 3*). Figure 9 illustrates the resulting Trustworthy AI-based Big Data Management (TAI-BDM) Reference Model - a combination of Kaufmann’s refined BDM Reference Model with the AIGO’s high-level structure of a generic AI system.

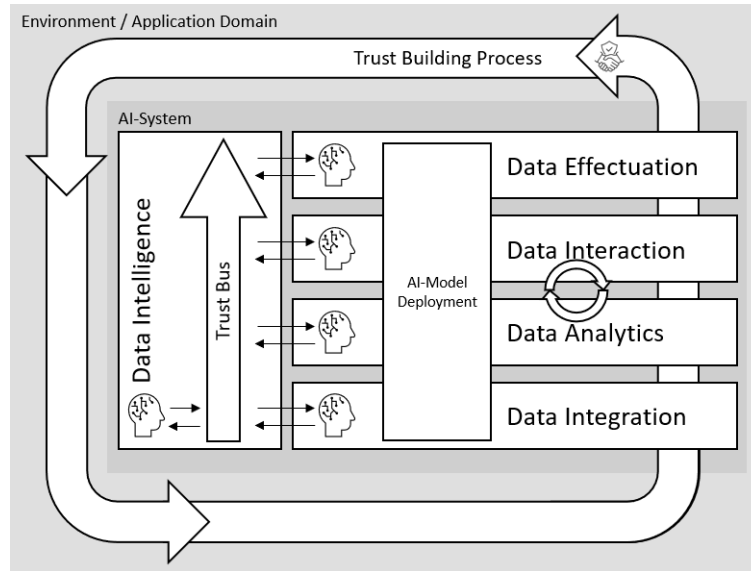


Fig. 9. Trustworthy AI-based Big Data Management (TAI-BDM) Reference Model

Revisiting AIGO’s system description, whereas “*the key power of an AI system resides in its operational logic*” [37], its outputs “*take the form of recommendations, predictions or decisions that can influence the state of the environment*” [37]. In this way, the inner part of Kaufmann’s refined BDM Reference Model (AI-supported *data integration*, *data analytics*, *data interaction*, *data effectuation*, and *data intelligence* layer) represents the *AI system* according to AIGO’s definition, the *AI-model deployment* layer represents its *operation logic*, and the outer loop including the *trust building process* between individuals and the system itself represents the *environment / application domain* layer.

Concluding this conceptual modeling and design section of trustworthiness in AI-based big data analysis and following the general structure of this research, the conceptual modeling of trustworthiness in AI-based big data analysis (*RQ1*) has been outlined first. Afterwards, the integration of trustworthy artificial intelligence into the whole big data analysis process (*RQ2*) as well as the integration of trustworthy artificial intelligence in real-world productive big data analysis application scenarios (*RQ3*) has been modeled and discussed. In this way, the resulting TAI-BDM Reference Model extends Kaufmann’s “*emergent knowledge process of higher order*” [30] between the AI system and its environment to an emergent knowledge and trust building process of higher order.

4 Discussion and Outlook

The benefit and promise of AI for science, industry, and society are undeniable. On the one hand, AI is helping people make better predictions and informed decisions, enabling innovation and productivity gains [31]. On the other hand, the risks and challenges that AI poses, including codifying and reinforcing unfair biases and infringing on human rights such as privacy, are also undeniable [31].

Therefore and based on the motivation statements 1-3 (*modeling of trustworthy artificial intelligence, trustworthy AI-support as a necessary requirement for modern (big data) analysis applications as well as the need to deploy and demonstrate their applicability and impact in productive industrial application solutions*), the contributions summarized in this research focus on ***trustworthiness in AI-based big data analysis***. Motivated by the importance that human users need opportunities to understand, to comprehend, and therefore to trust and to deploy AI applications successfully into production, this research has been guided by the research questions how trustworthiness of artificial intelligence in big data analysis can be modeled and qualified (*RQ1*), where and how trustworthy artificial intelligence can be integrated into the whole big data analysis process (*RQ2*), and how trustworthy artificial intelligence can be integrated in real-world productive big data analysis application scenarios (*RQ3*).

With focus on the first research question, the definition of intelligence, artificial intelligence, trust, and trustworthiness have been introduced and discussed. Even though newer approaches from cognitive psychology (Guilford’s Structure of Intellect [24] or Gardener’s Theory of Multiple Intelligences [22]) lead to an expansion of the general concepts of intelligence (c.f. Spearman [46] and Thurstone [51]). Nevertheless, intelligence is quite diverse and its expression can only be determined through natural observation and assessment of an individual in different real-world scenarios, which is important when defining and also evaluating artificial intelligence.

Since the early 1950s, “*when Alan Turing first posed the question of whether machines can think*” [37] [54], the artificial intelligence technical landscape has evolved significantly [37]. Initially defined as “*the science and engineering of making intelligent machines*” [34], “*AI has evolved from symbolic AI where humans built logic-based systems, through the AI ‘winter’ of the 1970s to the chess-playing computer Deep Blue in the 1990s*” [37]. In context to an AI system, the AI Group of Experts at the OECD developed in 2018 a description of an AI system “*that aims to be understandable, technically accurate, technology-neutral, and applicable to short- and long-term time horizons*” [37], which also helps to emphasize the importance of trust and trustworthiness.

Nevertheless, “*if AI systems do not prove to be worthy of trust, their widespread acceptance and adoption will be hindered, and the potentially vast societal and economic benefits will not be fully realized*” [31]. In this way, trust as well as trustworthiness has been introduced and discussed next. Between individuals, trust is on the one hand based in terms of their *competence* and on the other hand based in terms of their *honesty* [43]. Therefore, trust and trustworthiness have at least both a knowledge component and a moral component [25]. With regard to initial rational definition of trust (reliability, truth, or ability), ability and truth can be replaced by competence and honesty, whereas reliability remains as important third attribute according to the interpretation of Easton’s argumentation. Even if these dimensions of trust are suitable for individuals, they are inappropriate for describing a system. Therefore, corresponding dimensions (*reliability* to *reproducibility*, *honesty* to *validity*, and *competence* to *accuracy*) that are able to define trustworthiness based on the output of an individual or system instead of an individual or system itself [15] have been defined and discussed.

Based on these results, the dimensions of trust and trustworthiness in the interaction between individuals as well as between individuals and systems (c.f. Figure 7) addressing the modeling of trustworthiness in AI-based big data analysis (*RQ1*) have been introduced and discussed. Nevertheless, the implementation as well as the evaluation aspects have not yet been fully answered.

Moving forward to the second research question where and how trustworthy artificial intelligence can be integrated into the whole big data analysis process (*RQ2*), Kaufmann’s BDM Reference Model as well as Easton’s definition that trust based on the output of an individual or system instead of an individual or system itself [15] paved the way of modeling trustworthiness in AI-based big data analysis.

Kaufmann’s Reference Model for Big Data Management contains two different types of layers. Four technical layers on the right side (*data integration*, *data analytics*, *data interaction* and *data effectuation*) that improve the effective benefit of Big Data technologies to represent the technical big data analysis process as well as one cross-functional *data intelligence* layer “to ensure that knowledge and skills can be acquired and properly managed in the context of big data management” [30]. In this way, the integration of the technical aspects of artificial intelligence into the whole big data analysis process is located on the right side, whereas the integration of the aspects regarding to trustworthiness is located on the left side within the *data intelligence* layer.

With focus on the right side of the conceptual reference model for trustworthiness in AI-based big data analysis, technical aspects of artificial intelligence into the whole big data analysis process are only located within the first three layers *data integration*, *data analytics*, *data interaction*. The fourth *data effectuation* layer focuses “on the utilization of the data analysis results to create value in products, services, and operations of the organization” [30]. Therefore, artificial intelligence does not play a role in this layer. The second *data analytics* layer represents the most important layer in context of artificial intelligence at first glance. Nevertheless, artificial intelligence also plays an important role within both *data integration* and *data interaction* layers.

Starting with *data integration*, artificial intelligence can be utilized within the combination process of data from different distributed and heterogeneous data sources into a single platform with consistent access [30]. *Data analytics* includes “the transformation of raw data into useable information” [30]. This layer represents the most important layer in context of AI because the application of data science that is defined as “the extraction of actionable knowledge directly from data through a process of discovery, or hypothesis formulation and hypothesis testing” [11] takes place in this layer. Whereas all AI-related layers can implement symbolic AI approaches, which use logical representations to deduce a conclusion from a set of constraints and require that researchers build detailed and human-understandable decision structures to translate real-world complexity and help machines arrive at human-like decisions [37], artificial intelligence within the *data analytics* implements primarily statistical AI approaches that allow machines to learn in an automated manner through patterns and inferences rather than through explicit instructions from a human [37]. Finally, *data interaction* consists of mutual interferences of users and data analysis results that generate individual and organizational knowledge, “which is important to note that data analysis results are in fact nothing but more data unless users interact with them” [30]. In this way, artificial intelligence can be utilized to support the generation and the enhancement of visualizations as well as to support the resulting human analysis process of the generated visualizations within the interaction between humans and the big data analysis results. Additionally, a cross-functional *AI-model deployment* layer is defined next to the small feedback loop between both *data analytics* and *data interaction* layers that highlights the iterative process of planning and designing, building and interpretation, verification and validation, as well as the deployment of statistical AI models into enterprise-wide production [37].

With focus on the left side of the conceptual reference model for trustworthiness in AI-based big data analysis, *data intelligence* refers to the ability of the organization to acquire knowledge and skills in the context of big data management [30]. Revisiting Easton’s definition that trust is based on the output of an individual or system instead of an individual or system itself [15] in combination with the integration of artificial intelligence within the right-sided layers of the conceptual reference model for trustworthiness in AI-based big data analysis, each layer with AI-support needs an individual trust-component based on its output within the *data intelligence* layer. Nevertheless, the overall trustworthiness of a system depends on the trustworthiness of the outcome of the whole process. Thus, the individual trustworthiness of the AI-supported layers needs to be integrated into an overall *trust building process*. In this process, the dimensions

of trustworthiness in the interaction between individuals and AI components within the certain layers as well as information about past interactions between individuals and the overall AI system within “*the emergent knowledge process of higher order*” [30], which is symbolized by the outer loop of the conceptual reference model for trustworthiness in AI-based big data analysis, need to be stored to gauge trustworthiness of the overall AI-based big data analysis to enable a trustworthy AI [1].

Based on these intermediate results, the integration of trustworthy artificial intelligence into the whole big data analysis process (*RQ2*) has been introduced, modeled, and discussed. Nevertheless, the overall research goal of modeling trustworthiness in AI-based big data analysis has not yet been fully answered. In this way, the implementation as well as the evaluation aspects of trustworthiness can be identified as remaining challenge of this research.

Finally, and with focus on the third research question how trustworthy artificial intelligence can be integrated in real-world productive big data analysis application scenarios (*RQ3*), AIGO’s high-level structure of a generic AI system (c.f. Figure 6) closed the gap between the modeling of a trustworthy AI-based big data analysis process and its interaction between with its environment within a real-world productive big data analysis application scenario.

Whereas the inner part of Kaufmann’s extended BDM Reference Model (AI-supported *data integration*, *data analytics*, *data interaction*, *data effectuation*, and *data intelligence* layer) represents the *AI system* according to AIGO’s definition, the *AI-model deployment* layer represents its *operation logic*, and the outer loop including the *trust building process* between individuals and the system itself represents the environment / application domain layer.

In this way, the resulting conceptual TAI-BDM Reference Model, which addresses the overall research goal of modeling trustworthiness in AI-based big data analysis, extends Kaufmann’s “*emergent knowledge process of higher order*” [30] between the AI system and its environment to an emergent knowledge and trust building process of higher order.

Nevertheless, the conceptual TAI-BDM Reference Model represents only a conceptual meta model. Moreover and due to the fact that Kaufmann’s original Reference Model for Big Data Management serves as a basis for IVIS4BigData, TAI-BDM also serves as a basis for the necessary adaptations of IVIS4BigData. Revisiting the outlined challenges of the application scenarios in science and industry (c.f. Section 1), the importance of trustworthy AI-supported data exploration in big data analysis applications has been highlighted. Unfortunately, IVIS4BigData does not meet these new requirements and must therefore also be adapted to the new circumstances in terms of trustworthy artificial intelligence.

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